

Floor Function Homework

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = \lfloor x \rfloor + \lfloor -x \rfloor + x.$$

What is $f(\pi)$?

- (A) $\pi - 2$ (B) π (C) $\pi - 1$ (D) $\pi + 1$ (E) $2\pi + 1$

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function given by

$$f(x) = x - \lfloor x \rfloor.$$

What is the range of the function f ?

- (A) $\emptyset$ (B) $\mathbb{N}$ (C) $\mathbb{Z}$ (D) $[0, 1)$ (E) $\mathbb{R}$

3. What is the set of the solution to the equation

$$\lfloor 3 - x \rfloor = -6?$$

- (A) $\mathbb{R}$ (B) $\{8, 9\}$ (C) $(8, 9]$ (D) $(8, 9)$ (E) $\emptyset$

4. How many integers x are there satisfying

$$\left\lfloor \frac{2x}{3} - \frac{1}{2} \right\rfloor = 3?$$

- (A) 5 (B) 4 (C) 3 (D) 2 (E) 1

5. Solve the equation

$$\lfloor x + 3 \rfloor + \lfloor x - 3 \rfloor = 4.$$

- (A) $-1 \leq x < 0$ (B) $0 \leq x < 1$ (C) $1 \leq x < 2$ (D) $2 \leq x < 3$ (E) $3 \leq x < 4$

6. How many **integers** x are there satisfying

$$\lfloor x \rfloor^2 - 6\lfloor x \rfloor + 5 = 0?$$

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

7. Solve the inequality

$$\lfloor 2x - 3 \rfloor \leq 4.$$

- (A) $x > \frac{3}{2}$ (B) $x \leq 4$ (C) $x < 4$ (D) $x \geq 4$ (E) $x \leq \frac{3}{2}$

8. What is the sum of the positive integers x satisfying

$$\lfloor 4 - x \rfloor \geq -2?$$

- (A) 18 (B) 21 (C) 24 (D) 27 (E) 30

9. What is the sum of the real numbers x satisfying

$$2x - \lfloor x \rfloor = 3?$$

- (A) $\frac{5}{2}$ (B) 3 (C) $\frac{7}{2}$ (D) $\frac{9}{2}$ (E) $\frac{11}{2}$

10. What is the set of the solution to the equation

$$\left\lfloor \frac{1}{x} \right\rfloor = 3$$

- (A) $[\frac{1}{3}, \frac{1}{2}]$ (B) $(\frac{1}{4}, \frac{1}{3}]$ (C) $[\frac{1}{3}, \frac{1}{2})$ (D) $[\frac{1}{4}, \frac{1}{3})$ (E) $[\frac{1}{5}, \frac{1}{4})$

11. Given that

$$f(x) = \left\lfloor \frac{5}{x} \right\rfloor + \left\lfloor \frac{x}{5} \right\rfloor$$

compute $f(1) + f(2) + f(3) + \dots + f(10)$.

- (A) 13 (B) 15 (C) 17 (D) 19 (E) 21

12. Compute

$$\prod_{k=1}^{16} \left\lfloor \frac{k+3}{k} \right\rfloor$$

- (A) 4 (B) 8 (C) 16 (D) 21 (E) 25

13. Compute

$$\sum_{k=1}^{100} \left\lfloor \frac{k}{10} \right\rfloor$$

- (A) 350 (B) 390 (C) 400 (D) 460 (E) 490

14. What is the largest domain the function

$$f(x) = \sqrt{\frac{x}{\lfloor x-1 \rfloor}}$$

can be defined on?

- (A) $\mathbb{R} \setminus \{1\}$ (B) $\mathbb{R} \setminus \{0, 2\}$ (C) $\mathbb{R} \setminus (0, 2)$ (D) $\mathbb{R} \setminus [1, 2)$ (E) $\mathbb{R} \setminus (-1, 1)$

15. What is the set of the solution to the equation

$$\lfloor 2x \rfloor = \lfloor x - 2 \rfloor?$$

- (A) $[-\frac{7}{2}, -\frac{5}{2})$ (B) $[-\frac{7}{2}, -\frac{1}{2})$ (C) $[-\frac{5}{2}, -\frac{3}{2})$ (D) $[-\frac{5}{2}, \frac{1}{2})$ (E) $[-\frac{3}{2}, \frac{1}{2})$

16. How many positive integers k are there satisfying

$$\left\lfloor \frac{k}{5} \right\rfloor = \left\lfloor \frac{k}{6} \right\rfloor$$

17. What is the sum of the real numbers x satisfying

$$\left\lfloor \frac{6x+5}{8} \right\rfloor = \frac{15x-7}{5}?$$

18. Determine the number of real solutions of

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{2x}{3} \right\rfloor = x$$

19. Find all solutions to the equation

$$x^2 + 7 = 8\lfloor x \rfloor.$$

20. Solve the equation

$$\lfloor x \rfloor + \lfloor x^2 \rfloor + \lfloor x^3 \rfloor = \{x\} - 1.$$

Hint: The left side of the equation is an integer, so the right side must be an integer as well.

21. Solve the system of equations

$$x + \lfloor y \rfloor + \{z\} = 13.2$$

$$y + \lfloor z \rfloor + \{x\} = 15.1$$

$$z + \lfloor x \rfloor + \{y\} = 14.3$$

22. Find one real number x satisfying

$$\lfloor x \rfloor + \lfloor 2x \rfloor + \lfloor 4x \rfloor = 123.$$

Hint: $\frac{123}{7}$ is approximately 17.5, we should find something close to that.

23. (2020 AMC 10A #22) For how many positive integers $n \leq 1000$ is

$$\left\lfloor \frac{998}{n} \right\rfloor + \left\lfloor \frac{999}{n} \right\rfloor + \left\lfloor \frac{1000}{n} \right\rfloor$$

not divisible by 3? (Recall that $\lfloor x \rfloor$ is the greatest integer less than or equal to x .)

(A) 22 (B) 23 (C) 24 (D) 25 (E) 26

Hint: If all of the three terms are equal to each other, then their sum is clearly divisible by 3 and we do not want that. When is $\left\lfloor \frac{999}{n} \right\rfloor$ different than $\left\lfloor \frac{998}{n} \right\rfloor$?

24. (2021 Fall AMC 10A #16) The graph of

$$f(x) = |\lfloor x \rfloor| - |\lfloor 1 - x \rfloor|$$

is symmetric about which of the following? (Here $\lfloor x \rfloor$ is the greatest integer not exceeding x .)

(A) the y -axis (B) the line $x = 1$ (C) the origin (D) the point $\left(\frac{1}{2}, 0\right)$ (E) the point $(1, 0)$

Note: In the definition of the function, we take absolute value after floor function. Try drawing the graph.

25. Find the smallest positive integer n such that there exists an integer x satisfying

$$\left\lfloor \frac{10^n}{x} \right\rfloor = 2026$$

Hint: Write an inequality to find a range for x in terms of n . Also note that $\frac{1}{2026} = 0.00049358\dots$ and $\frac{1}{2027} = 0.00049333\dots$